

Disproof of Conjecture 00000000297: no set of two Wang tiles is aperiodic

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Abstract

Conjecture 00000000297 asserts that there is an aperiodic rotation-free set of only 2 Wang tiles, that no single Wang tile is aperiodic, and hence that the minimal size of a rotation-free aperiodic Wang tile set is exactly 2. We prove that every set of at most two Wang tiles that tiles the plane also tiles it with periods $(2, 0)$ and $(0, 2)$. Hence no set of at most two Wang tiles is aperiodic, in the strong sense (no tiling has a nonzero period) or in the weak sense (no tiling is doubly periodic). The first clause of the conjecture, and with it the claim that the minimum is exactly 2, is false. The proof is a short case analysis. It is formalized in Lean 4 (core library only), and an independent Python script checks it by exhaustive search.

1 Statement and reading

The conjecture (English) reads:

Definition: Wang tiles are unit squares tiled by matching edge colors only. Conjecture: There exists an aperiodic prototile set of only 2 tiles without rotations, and no single rotation-free aperiodic tile exists (the minimal rotation-free aperiodic set is exactly 2).

The Chinese version says the same. It is a conjunction of three claims:

(C1) there is an aperiodic set of only 2 Wang tiles (without rotations);

(C2) no single Wang tile is aperiodic;

(C3) the minimal size of an aperiodic set of Wang tiles is exactly 2.

We show that (C1) is false. Since (C3) contains (C1), (C3) is false too, and so is the conjecture. (C2) is true; it is the one-tile case of our theorem.

Definition 1. Let $\mathcal{C}$ be any set of colors. A *Wang tile* is a quadruple $t = (n_t, e_t, s_t, w_t) \in \mathcal{C}^4$ (north, east, south and west edge colors). Tiles are placed on the cells of $\mathbb{Z}^2$ and are never rotated or reflected. A *tiling* of the plane by a finite tile set T is a map $f : \mathbb{Z}^2 \rightarrow T$ such that for all (x, y)

$$e_{f(x,y)} = w_{f(x+1,y)} \quad \text{and} \quad n_{f(x,y)} = s_{f(x,y+1)}.$$

Write $a \mid b$ (“ b may sit right of a ”) when $e_a = w_b$, and $a \perp b$ (“ b may sit above a ”) when $n_a = s_b$. A vector $(p, q) \in \mathbb{Z}^2$ is a *period* of f if $f(x+p, y+q) = f(x, y)$ for all x, y . The tiling f is *periodic* if it has a nonzero period, and *doubly periodic* if it has two linearly independent periods.

Definition 2. A tile set T is (*strongly*) *aperiodic* if it tiles the plane but no tiling by T is periodic. It is *weakly aperiodic* if it tiles the plane but no tiling by T is doubly periodic.

Strong aperiodicity is the standard notion for Wang tiles (Berger, Robinson, Jeandel–Rao), and it implies weak aperiodicity. (It is a classical fact that the two notions coincide for Wang tiles. We do not use it.) We refute (C1) under *both* readings. So that every reading is covered:

- “only 2 tiles” may mean exactly 2 or at most 2. We cover every set of at most 2 tiles, and allow the two tiles to be equal.
- The colors are arbitrary, from any color set $\mathcal{C}$, with any number of colors.
- “Without rotations” is part of the definition of Wang tiles: tiles are placed by translation only. This is the setting of the conjecture.

The conjecture’s definition fixes the setting: Wang tiles, i.e. edge-colored unit squares. We say nothing about general shapes (“prototiles” that are polygons or other sets). That is a different question.

“Aperiodic prototile set” is the standard term of art (Grünbaum–Shephard): the set tiles the plane, and every tiling by it is non-periodic. A set that merely *admits* some non-periodic tiling is not aperiodic in this sense. For example, the tiles $A = (n, e, s, w) = (0, 1, 0, 1)$ and $B = (0, 2, 0, 2)$ force constant rows, but the rows can be stacked in any order, so non-periodic tilings exist. That loose reading would make (C1) true, and we set it aside as non-standard.

2 The theorem

Theorem 3. *Let T be a set of at most two Wang tiles. If T tiles the plane, then some tiling by T has the periods $(2, 0)$ and $(0, 2)$. In particular it is periodic and doubly periodic, so T is neither aperiodic nor weakly aperiodic.*

If $T = \emptyset$ there is no tiling. If $T = \{t\}$, put $t_0 = t_1 = t$. So assume $T = \{t_0, t_1\}$ (possibly $t_0 = t_1$) and let f be a tiling by T . Call the pair *H-crossing* if $t_0 \mid t_1$ and $t_1 \mid t_0$, and *V-crossing* if $t_0 \perp t_1$ and $t_1 \perp t_0$.

Lemma 4 (local lemma). *If $\{t_0, t_1\}$ is not H-crossing, then $f(x, y) \mid f(x, y)$ for every cell (x, y) . If it is not V-crossing, then $f(x, y) \perp f(x, y)$ for every cell.*

Proof. Let $a = f(x-1, y)$, $b = f(x, y)$, $c = f(x+1, y)$. Then $a \mid b$ and $b \mid c$. Suppose $b \mid b$ fails. Then $c \neq b$, since $b \mid c$. Also $a \neq b$, since $a \mid b$. As $a, b, c \in \{t_0, t_1\}$, this forces $a = c$ to be the tile other than b . Then $a \mid b$ and $b \mid a$ are the two cross matches, so the pair is H-crossing, a contradiction. The vertical statement is proved the same way, with $f(x, y-1)$, $f(x, y)$, $f(x, y+1)$. $\square$

Proof of Theorem 3. Each pattern g below takes values in $\{t_0, t_1\}$ and depends only on $x \bmod 2$ and $y \bmod 2$, so it has periods $(2, 0)$ and $(0, 2)$. These are linearly independent ($\det \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 4 \neq 0$), and $(2, 0) \neq (0, 0)$. It remains to check that each g is a tiling.

Case 1: H-crossing and V-crossing. Take the checkerboard $g(x, y) = t_0$ if $x + y$ is even and t_1 otherwise. Horizontal and vertical neighbours are always t_0, t_1 or t_1, t_0 , in that order. These pairs match by assumption.

Case 2: neither. By Lemma 4, the tile $t = f(0, 0)$ satisfies $t \mid t$ and $t \perp t$. The constant pattern $g \equiv t$ is a tiling.

Case 3: not H-crossing, but V-crossing. By Lemma 4, every tile used by f matches itself horizontally. If $f(0,1) = f(0,0) = t$, then $t \mid t$ by the lemma, and $t \perp t$ because $f(0,0) \perp f(0,1)$. So $g \equiv t$ is a tiling. Otherwise $f(0,0) \neq f(0,1)$ are the two tiles t_0, t_1 in some order. Hence $t_0 \mid t_0$ and $t_1 \mid t_1$. Take horizontal stripes: $g(x,y) = t_0$ if y is even, t_1 otherwise. Horizontal neighbours are equal and match. Vertical neighbours are t_0 below t_1 , or t_1 below t_0 , and these pairs match by V-crossing.

Case 4: H-crossing, but not V-crossing. This is Case 3 with the axes swapped. Use $f(0,0)$ and $f(1,0)$, and the vertical stripes $g(x,y) = t_0$ if x is even, t_1 otherwise. If $t_0 = t_1$ the pair can still be H- or V-crossing. The argument does not use $t_0 \neq t_1$. $\square$

Corollary 5. *No set of at most two Wang tiles is aperiodic, even weakly. Hence:*

1. (C1) is false. There is no aperiodic set of 2 Wang tiles, whether “2” means exactly or at most 2, and whether “aperiodic” is strong or weak.
2. (C2) is true: a single tile is never aperiodic.
3. (C3) is false: the minimal size of an aperiodic Wang tile set is not 2.

In particular the conjecture, a conjunction containing (C1), is false.

Remark 6. This agrees with the literature. Jeandel and Rao (2021) found an aperiodic set of 11 Wang tiles. They also showed by computer search that every Wang set with at most 10 tiles is not aperiodic. So the true minimum is 11. Our argument for two tiles is elementary and does not use their result.

Remark 7 (Why the local lemma looks at three cells). A 2×2 window is not enough to decide the question. Take $t_0 = (0, 1, 0, 2)$ and $t_1 = (1, 3, 1, 1)$, written as (n, e, s, w) . The only horizontal match is $t_0 \mid t_1$, and the only vertical matches are $t_0 \perp t_0$ and $t_1 \perp t_1$. The 2×2 square with rows $t_0 t_1$ is legal. But no row of three cells is legal, so the pair does not tile the plane, and no periodic pattern exists. Lemma 4 looks at three consecutive cells $x-1, x, x+1$ of a row (or column). The script `verify.py` finds 16 relation pairs of this kind, all of which have a legal 2×2 square but no legal 3×3 square.

3 Lean formalization

The folder `lean4/` is a Lean 4.19.0 project that uses only the core library. Everything is defined from scratch in `Main.lean`:

- **Tile C:** a structure with fields `n e s w : C`, for an arbitrary color type `C`. The relations `HMatch a b := a.e = b.w` and `VMatch a b := a.n = b.s`.
- **IsTiling T f:** `f : Int → Int → Tile C` with `f x y ∈ T`, `HMatch (f x y) (f (x+1) y)` and `VMatch (f x y) (f x (y+1))` for all `x y`. Tile sets are lists, and `Tiles T := ∃ f, IsTiling T f`.
- **IsPeriodic f:** some period $(a,b) \neq (0,0)$. **IsDoublyPeriodic f:** periods (a,b) and (c,d) with $a*d - b*c \neq 0$.
- **Aperiodic T := Tiles T ∧ ∀ f, IsTiling T f → ¬ IsPeriodic f,** and **WeaklyAperiodic T** is the same with **IsDoublyPeriodic**. `Aperiodic.weakly` proves that strong implies weak.

The main results:

- `hloop`, `vloop`: Lemma 4.
- `two_tiles_periodic`: the four cases of the proof, with the patterns `constPat`, `checker`, `stripesH` and `stripesV`.
- `le_two_tiles_periodic`: for every list `T` with `T.length ≤ 2`, `Tiles T` implies a tiling with periods $(2, 0)$ and $(0, 2)$ that is periodic and doubly periodic. Lists of length 0, 1 and 2 are all covered.
- `not_aperiodic`, `not_weaklyAperiodic`: no list of length at most 2 is (weakly) aperiodic.
- `Conjecture297 C := Clause1 C ∧ Clause2 C ∧ MinAperiodicSize C 2`, where `Clause1 C := ∃ T, T.length = 2 ∧ Aperiodic T`. `conjecture_00000000297_false : ¬ Conjecture297 C` holds for every color type `C`. The project also proves `clause1_false`, `clause1_weak_false` (weak reading), `clause1_le_false` (“at most 2” with the weak reading), `minSize_ne_two`, and `clause2_true`.

Non-vacuity checks, with natural-number colors:

- `tiles_example`: a 2-tile set that tiles.
- `not_tiles_example`: a 2-tile set that does not tile.
- `crossTiling_isTiling` and `crossTiling_not_periodic`: a four-tile set (blank, horizontal line, vertical line, cross) with a tiling that has a single cross at the origin. This tiling has no nonzero period. So `IsPeriodic` is a real restriction on tilings, and our theorem produces *some* periodic tiling. It does not claim that every tiling is periodic.

The project has no `sorry`, `native_decide` or custom axioms. `#print axioms` shows only `propext`, `Classical.choice` and `Quot.sound`.

4 Independent check

`verify.py` (Python 3 standard library) uses a different method. It abstracts a pair of tiles $\{0, 1\}$ to its two compatibility relations $H, V \subseteq \{0, 1\}^2$. For all $16 \times 16 = 256$ pairs (H, V) , including pairs that no coloring realises, it checks by brute force that either no 3×3 square can be filled legally, or a legal 2×2 torus pattern exists. Every tiling of the plane restricts to a legal 3×3 square, and a 2×2 torus pattern gives a tiling with periods $(2, 0)$, $(0, 2)$. So this reproves Theorem 3. There are 127 pairs with a torus and 129 pairs with no legal 3×3 square. The script also runs the test on all $3^8 = 6561$ ordered pairs of concrete tiles with colors in $\{0, 1, 2\}$, which realise 144 distinct relation pairs. It confirms that 3×3 cannot be replaced by 2×2 (16 relation pairs have a legal 2×2 square but no torus). Finally, it checks that the four patterns of the proof always suffice.

5 Limitations

Everything above is proved in Lean, for an arbitrary color type, with tile sets as lists of length at most 2. The literature remark (Jeandel–Rao) is context only and is not used. We do not address prototiles that are general shapes. That setting is different. Greenfeld and Tao (*Undecidable translational tilings with only two tiles, or one nonabelian tile*, Discrete Comput. Geom. 2023)

construct an aperiodic pair of translational tiles in $\mathbb{Z}^2 \times G_0$ with G_0 finite abelian. For a single translational tile of $\mathbb{Z}^2$, or a polygonal topological disk, periodicity is known (Bhattacharya; Girault-Beauquier–Nivat, Kenyon). The conjecture’s own definition (“Wang tiles are unit squares tiled by matching edge colors only”) fixes the Wang setting, which is the one refuted here.